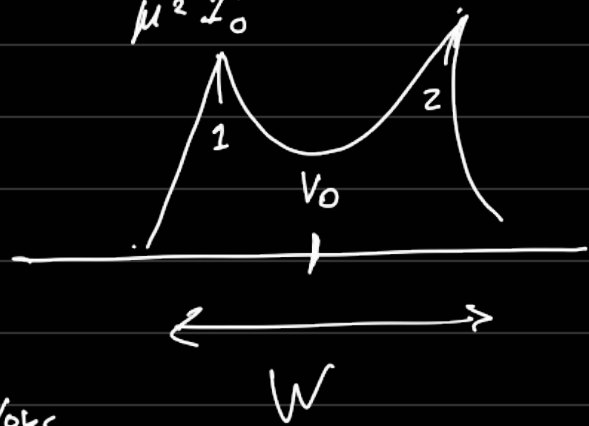


* B.2.1 (b) Tully-Fisher

$$L \propto \frac{V^4}{\mu^2 I_0}$$

(c) $W = 2 V_{max} \sin(i)$ *

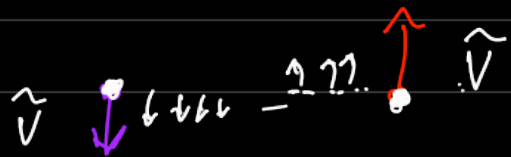


$$(V_{obs})_1 = 600 \text{ km/s}$$

$$(V_{obs})_2 = 1000 \text{ km/s}$$



$$\sin(i) = \frac{V_{obs}}{V_{true}}$$



$$V_{obs} = \sin(i) V_{true}$$

$\xleftarrow{600} \quad \xrightarrow{1000}$
 $\xleftarrow{800}$

← from rel. motion

$$\begin{cases} (V_{obs})_2 = V_s + \tilde{V} \\ (V_{obs})_1 = V_s - \tilde{V} \end{cases}$$

$$\begin{cases} (V_{obs})_1 + (V_{obs})_2 = 2V_s \Rightarrow V_s = \frac{1}{2}(1600) = \underline{800} \\ (V_{obs})_2 - (V_{obs})_1 = 2\tilde{V} \Rightarrow \tilde{V} = \underline{200} \end{cases}$$

$V_s \approx 800 \text{ km/s}$ *

* $W = ((V_{true})_2 - (V_{true})_1) = 2 V_{max} \sin(i)$

$$V_{true} = \frac{V_{obs}}{\sin(i)}$$

Plug in

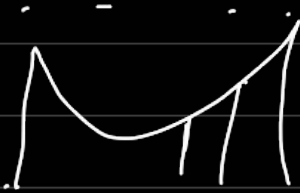
$$((V_{obs})_2 - (V_{obs})_1) / \sin(i) = 2 V_{max} \sin(i)$$

$$(V_{obs})_2 - (V_{obs})_1 = 2 V_{max} \sin^2(i)$$

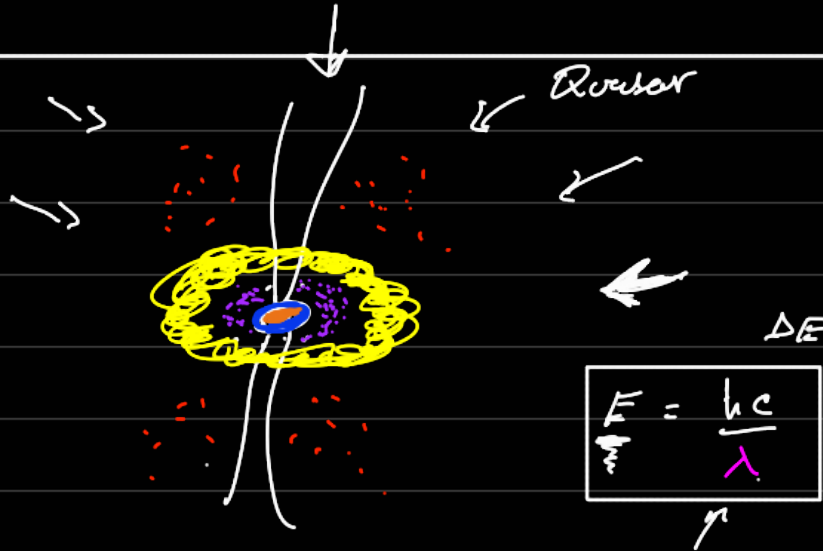
$$V_{max} = \left(\frac{1}{2 \sin^2(i)} \right) ((V_{obs})_2 - (V_{obs})_1)$$

$$* \text{ Shift } = \frac{b}{a}$$

$$* \boxed{z = \frac{b}{a}} *$$



AGN:



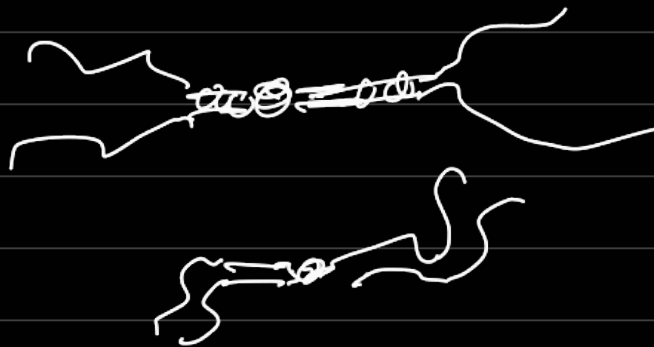
- Broad: X-rays
- Narrow: $\Delta E_N \ll \Delta E_B$

* AGNs: looking at the Gal. from a different angle
 $\rightarrow$ $\left[\begin{array}{l} \text{FR I} \\ \text{FR II} \end{array} \right]$

FR II:



FR I:



* Alfven. waves / Flux freezing:

Black hole:

evidence for a B.H.?

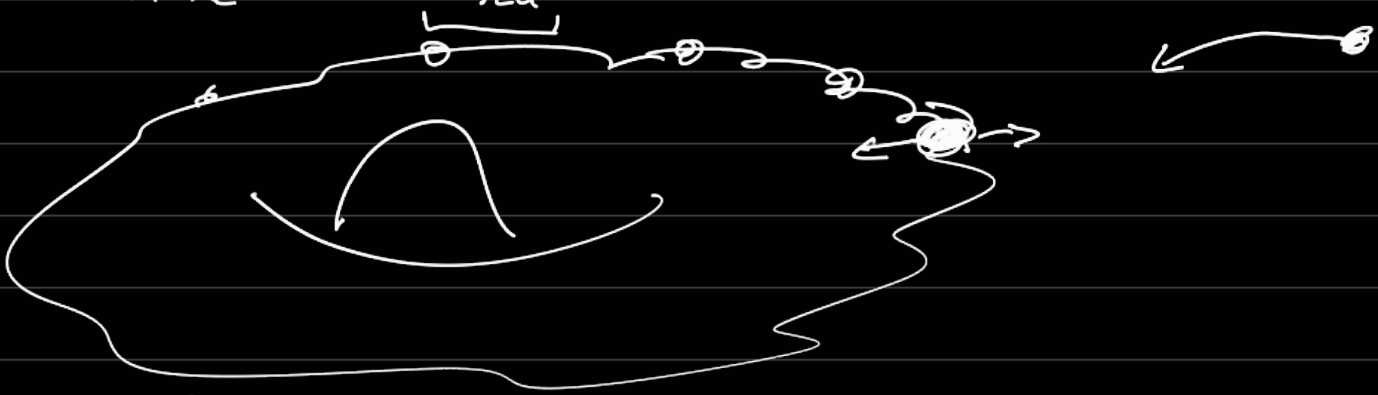
- Nuclear or supernova: Not enough m. ←
- Stellar motion near galactic center: eccentric
→ Very high mass to be concentrated.

$$M_{BH} = M_{BH}(0) e^{kt}$$

→ Newtonian: $\frac{1}{2} m c^2 \rightarrow \frac{G M_{BH} m}{R_{Sch}} = 0$

$$R_{Sch} = \frac{2 G M_{BH}}{c^2}$$

$$R_{stable} = 3 R_{Sch}$$



Equilibrium:



Assumptions: (1) BH is central → Basically a point mass
(2): e^- and p^+ will have to move together

$$(1) \Rightarrow F_g = \frac{G M_{BH} \tilde{m}}{r^2} \leftarrow \cancel{m_e}$$

$$(2) \quad \tilde{m} = (m_e + m_p) \leftarrow m_H$$

$$((1)+(2)) \rightarrow F_g \approx \frac{G M_B m_p}{r^2}$$

Finite B.H.:

$$F_r = \frac{dp}{dt}$$



$$E = pc$$

$$F = \frac{dp}{dt} = \frac{d(mv)}{dt} = m \frac{dv}{dt}$$

$$F_r = \frac{1}{c} \frac{dE}{dt}$$

$$L = \frac{dE}{dt}$$

$$(F_r)_{\text{tot}} = \frac{L}{c}$$



→ Force only along the $(m_p + m_e)$:

$$(F_r) = \left(\frac{\sigma_T}{4\pi R^2} \right) (F_r)_{\text{tot}} = \frac{\sigma_T}{4\pi R^2} \frac{L}{c}$$

$$F_r = F_g \Leftrightarrow \frac{\sigma_T L}{4\pi R^2 c} = \frac{G M_{BH} m_p}{R^2}$$

$$L_{\text{edd}} = \left(\frac{4\pi c G m_p}{\sigma_T} \right) M_{BH}$$

$$L_{\text{edd}} = 1.26 \times 10^{31} \left(\frac{M_{BH}}{M_\odot} \right)$$

$$* E_0 = mc^2 \leftarrow$$

$$* E = \epsilon mc^2 *$$

$$L = \frac{dE}{dt} = \epsilon c^2 \frac{dm}{dt} = \epsilon c^2 \dot{m}$$

$$\Rightarrow \frac{dm}{dt} = () \dot{M}_{BH}$$

ϵ : Radiative eff.



$$\dot{M}_{BH} = \underbrace{\dot{m}}_{\text{total}} - \underbrace{\epsilon \dot{m}}_{\text{lose}} = \dot{m}(1-\epsilon)$$

$$(\dot{m} - \epsilon \dot{m})$$

$$\text{Plug in: } L = \epsilon c^2 \dot{m} = \epsilon c^2 \left(\frac{\dot{M}_{BH}}{(1-\epsilon)} \right)$$

$$L = \frac{4\pi c G m_p}{\sigma_T} M_{BH} = \frac{\epsilon c^2}{1-\epsilon} \dot{M}_{BH}$$

$$L_{\text{edd}} = \frac{\epsilon c^2 \sigma_T}{(1-\epsilon) 4\pi c G m_p}$$

$$* \epsilon = 0.1 *$$

$$\int \frac{1}{\dot{m}} dt = \int \frac{dm}{\dot{m}}$$

$$M_{BH} = t_{edd} \dot{M} \Rightarrow \int_{t_{edd}}^{t} \dot{M} dt = M - M_0$$

$$c + \frac{t}{t_{edd}} = \ln(M) \Rightarrow M(t) = e^c e^{t/t_{edd}}$$

$$e^c \equiv M_0$$

$$* \boxed{M_{BH}(t) = M_0 e^{t/t_{edd}}} *$$

$$k = \frac{1}{t_{edd}}$$

Notes: $M_{BH}(t) = M_0 e^{kt}$

$$\rightarrow M_{BH} = t_{edd} \frac{dM}{dt}$$

$$I(R) = I_0 e^{-R/k_H} \quad R_H:$$

t_{edd} : time for BH to grow by a factor of e

$$k = (\dots) s^{-1} \Rightarrow t_{edd} \approx \frac{1}{k} = \underline{\underline{4.5 \times 10^7 \text{ yrs}}}$$

$$c = 0.1$$

$$M_{BH}(t) = M_0 e^{t/t_{edd}}$$

$$M_{BH}(t_{edd}) = M_0 e^1 = \frac{e}{2} M_0$$

$$[O_T] = m^2$$

$$M_{BH}(t_p) = 2M_0 \Rightarrow 2M_0 = M_0 e^{t_p/t_{edd}}$$

$$\boxed{t_{edd} \ln(2) = t_p}$$

Crossing time:



$$\boxed{R = \dots}$$

$$\boxed{R = \frac{c \Delta t}{2}} \Rightarrow R = R_s$$

$$\Delta t \approx \frac{R_s}{c} = \frac{2GM_{BH}}{c^3}, \quad M_{BH} = 10^8 M_\odot$$

* $3R_s$: stable orbit

$R_s \leq R < 3R_s$: Can still orbit but might be unstable

Density functions:

IMF: $\int \xi(m) dm = ?$
(Stars in a galaxy)

$$[\xi(m) dm]$$

= # Stars with masses
m to (m + dm)

(i) $\int \xi(m) dm = N \leftarrow$

(ii) $\int_{m_1}^{m_2} \xi(m) dm = \underline{M_{tot}} (m_1 \text{ to } m_2)$

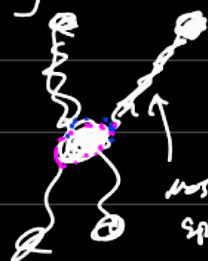
$$\xi(m) = \xi_0 \left(\frac{m}{M_\odot} \right)^{-\alpha} \leftarrow 1.7$$

Luminosity function: $[\phi(L) dL] =$ # of galaxies with L_{lum}
between L and L + dL
pr unit volume

* $\int_{L_0}^{L_1} \phi(L) dL = \frac{\# \text{ of gal}}{\text{Vol}} \text{ (in a region)}$

* $\int_{L_0}^{L_1} L \phi(L) dL = \frac{L_{lum} (L_0 \text{ to } L_1)}{\text{Vol}}$

* Looking over our notes on Cosmological scales



• DM densities: $\gg$ Rod. matter

• Central regions: Brightest $\rightarrow$ mostly elliptical gal.

Spirals only at edges

MW:



low redshift: * $V_{rad} = D H_0 + V_{pec}$

* $\boxed{V = cz}$ (rec. vel.)

